

Model Comparison — Holdout 2022-2025
RMSE, Skill vs RW, R², Hit Rate

Horizon	CatBoost RMSE	CatBoost RW RMSE	CatBoost Skill %	CatBoost R²	CatBoost Hit	XGBoost RMSE	XGBoost RW RMSE	XGBoost Skill %	XGBoost R²	XGBoost Hit	HTBoost RMSE	HTBoost RW RMSE	HTBoost Skill %	HTBoost R²	HTBoost Hit	Lasso RMSE	Lasso RW RMSE	Lasso Skill %	Lasso R²	Lasso Hit	SARIMA RMSE	SARIMA RW RMSE	SARIMA Skill %	SARIMA R²	SARIMA Hit	OLS RMSE	OLS RW RMSE	OLS Skill %	OLS R²	OLS Hit
0w	0.0656	0.0714	+8.2%	0.156	64.9%	0.0672	0.0714	+5.8%	0.113	59.9%	0.0602	0.0714	+15.7%	0.288	67.1%	0.0671	0.0735	+8.7%	0.164	63.3%	0.0673	0.0714	+5.7%	0.110	61.7%	—	—	—	—	—
1w	0.1006	0.1142	+11.9%	0.224	70.1%	0.1024	0.1142	+10.4%	0.196	66.5%	0.0977	0.1142	+14.4%	0.267	66.5%	0.1019	0.1164	+12.4%	0.230	63.9%	0.1105	0.1142	+3.2%	0.063	57.5%	—	—	—	—	—
2w	0.1185	0.1375	+13.8%	0.257	71.4%	0.1198	0.1375	+12.9%	0.240	68.6%	0.1149	0.1375	+16.5%	0.302	67.3%	0.1196	0.1387	+13.8%	0.251	64.8%	0.1322	0.1375	+3.8%	0.074	65.9%	—	—	—	—	—
1m	0.1420	0.1690	+16.0%	0.293	73.9%	0.1420	0.1690	+16.0%	0.293	68.8%	0.1412	0.1690	+16.4%	0.301	72.5%	0.1400	0.1688	+17.1%	0.304	71.8%	0.1614	0.1690	+4.5%	0.086	61.0%	0.1399	0.1690	+17.2%	0.313	67.4%
3m	0.2527	0.3029	+16.6%	0.304	72.2%	0.2610	0.3029	+13.8%	0.257	66.0%	0.2276	0.3029	+24.9%	0.435	75.1%	0.2190	0.2930	+25.3%	0.436	74.4%	0.2750	0.3029	+9.2%	0.175	77.5%	0.2156	0.3015	+28.5%	0.488	75.5%
6m	0.2771	0.3631	+23.7%	0.417	75.5%	0.2744	0.3631	+24.4%	0.428	78.1%	0.2412	0.3631	+33.6%	0.558	87.2%	0.2311	0.3600	+35.8%	0.586	81.9%	0.3187	0.3631	+12.2%	0.229	82.7%	0.2264	0.3728	+39.3%	0.629	86.8%
12m	0.1996	0.2137	+6.6%	0.120	65.9%	0.2248	0.2137	-5.2%	-0.116	61.2%	0.2062	0.2137	+3.5%	0.061	64.1%	0.1618	0.1699	+4.8%	-0.003	62.8%	0.2080	0.2137	+2.6%	0.044	53.5%	0.2189	0.1692	-29.4%	-0.850	56.9%